Power Connector

*note order of pins
J1
GNDA
F1 FH1
F2 FH2
F3 FH3
−15V Supply
+15V Supply
+5V Supply
D10 D11 D12
Reverse Polarity Protection

Digital Control Signals

3.3V to Glasgow
0.1 uF C68
4.7u C73
+3.3V
GNDD GNDD
J2
Power Good
S.LATCH_X
S.LATCH_Y
S.WRITE_A
S.LATCH_A
S.CLK_D
S.CLK_A
From LVDS Adapter

S[LATCH_X LATCH_Y LATCH_A WRITE_A CLK_D CLK_A]

5V Filtering

FB4 LI0603G221R-10
Digital
TP_D3.1
+5VD
C2 10u
C16 0.1 uF
D2 LED
R3 3.3k
FB1 LI0603G221R-10
+5V Supply
TP_A3.1
+5VA
C1 10u
C12 0.1 uF
GNDA1
D9 LED
R89 3.3k
Analog

15V Filtering

D3 LED
R7 1.5k
FB2
TP+1
+15V Supply
+15V
C11 10u
C13 0.1 uF
FB3
C46 10u
C15 0.1 uF
−15V Supply
−15V
TP−1
LED
D1
1.5k
R9
GNDA2

Digital Data In/Out

J3
D−1 D−2 D−3 D−4 D−5 D−6 D−7 D−8
J4
D−9 D−10 D−11 D−12 D−13 D−14 D−15
D−[1..15]

X Scan

S.LATCH_X
Pull down to enable outputs
GNDD
U4
SN74ALVCH16374DGGR
D−1…D−15 → 1Q1…1Q8 / 2Q1…2Q8 → X−1…X−8
GNDD6
X−[1..14]
+3.3V
GNDD
C27 0.1 uF C28 0.1 uF
S.CLK_D
0.1 uF 0.1 uF C35 C37
0.1 uF 0.1 uF C39 C41
+3.3V +3V0
TP_Clk_D2
U8 AD9744ARUZ
CLK D0 LSB D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 MSB
IOUTA IOUTB SLEEP MODE FS_ADJ REFIO REFLO AVDD DVDD AGND DGND
Connect to GND/DVCOM for binary, Connect to VDD for two's complement
TP_IA_X1
TP_IB_X1
JP4
TP_X_MSB1
C55 C−Opt
R32 R36
GNDA4
GNDA8
R45 499R
R39 301
R40 301
+14V0
Vocm
THS4151 U12
−14V0
R46 499R
TP_X_A1
TP_X_B1
R52 1K
R53 1K
R47 0
R50 0
R59 6.82k
R70
LT1363 U13
TP_X_Out1
R62 0
J5
Coaxial Con Amphenol 031−5538−10RFX
Independant singled ended output scale adjust.
TP_X_Offset1
R54 1K
R55 1K
U18 LT1363
Adjustable −14V0
RV2
R79 1.2k
20k
R82
R77 22k
1.3k
R78
SW2
Presets
GNDA
Offset voltage
Presets R30 2.32k R72 4.75k R74 R75 6.82k
SW1
RV1
Adjustable
Full scale 24V
Single 20V
Diff 12V
14.4V
DAC output scale adjust
2k = full range

Y Scan

S.LATCH_Y
Pull down to enable outputs
GNDD
GNDD7
U5
SN74ALVCH16374DGGR
D−1…D−15 → 1Q1…1Q8 / 2Q1…2Q8 → Y−1…Y−8
Y−[1..14]
+3.3V
GNDD
C26 0.1 uF
S.CLK_D
0.1 uF 0.1 uF C34 C36
0.1 uF 0.1 uF C38 C40
+3.3V +3V0
TP_Clk_D2
U7 AD9744ARUZ
CLK D0 LSB D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 MSB
IOUTA IOUTB SLEEP MODE FS_ADJ REFIO REFLO AVDD DVDD AGND DGND
Connect to GND/DVCOM for binary, Connect to VDD for two's complement
TP_IA_Y1
TP_IB_Y1
JP5
TP_Y_MSB1
C51 C−Opt
R31 R33
GNDA16
R41 499R
R37 301
R38 301
+14V0
Vocm
THS4151 U11
−14V0
R42 499R
TP_Y_A1
TP_Y_B1
R51 1K
R48 1K
R60 6.82k
R73
LT1363 U14
TP_Y_Out1
R63 0
J6
Coaxial Con Amphenol 031−5538−10RFX
TP_Y_Offset1
R54 1K
R55 1K
U17 LT1363
Adjustable −14V0
RV5
R68 1.2k
20k
R71
R65 22k
1.3k
R69
SW3
Presets
GNDA
Offset voltage
Offset voltage
Presets R29 2.32k R76 4.75k R80 R81 6.82k
SW4
RV6
Adjustable
Full output scale 20V
GNDA

Power Regulators

+5V +3.3V Digital

R86 3.3k
R85 499
PG3.3A
PG3.3D
R100 10k
Q4 MMBT3904TT1G
LED D38
GNDD8 GNDD9
+5VD
U19 LT3045
EN/UV PG IN IN IN SET ILIM GND GND GND OUT OUT OUTS PGFB
C14 4.7u
C17 4.7u
R83 10k
R84 1.50k
C20 10u
C10 100 uF
+3.3V
TP+3.2
R87 499
R88 100k
Programmable "power good" level set by resistors
Vset = Iset * Rset
see reference tables

+5V +3.0V

R14 3.3k
R6 499
PG3.3A
R99 10k
Q2 MMBT3904TT1G
LED D4
+5VA
U1 LT3045
C3 4.7u
C5 4.7u
R1 453k
R4 750k
C18 10u
C7 100 uF
+3V0
TP+3.1
R10 499
R11 100k
Programmable "power good" level set by resistors
Vset = Iset * Rset
see reference tables

+15V +14V

R15 3.3k
R8 10k
PG+14A
R23 10k
Q1 MMBT3904TT1G
LED D5
GNDD2 TestPoint
+15V
U2 LT3045
C4 4.7u
C6 4.7u
R2 340k
R5 1K
C19 10u
C8 100 uF
+14V0
TP+2
R12 499
R13 100k
Programmable "power good" level set by resistors
Vset = Iset * Rset
see reference tables

−15V −14V

+14V0
R21 13k
R18 10k
PG−14A
R98 10k
Q3 MMBT3904TT1G
LED D6
GNDA17
GNDA3
−15V
U3 LT3094_MSE
C21 4.7u
C22 4.7u
R16 340k
R17 1K
C25 10u
C9 100 uF
−14V0
TP−2
R19 499
R20 100k
Programmable "power good" level set by resistors
Vset = Iset * Rset
see reference tables

Power Good

+3.3V
U20A 7421
PG−14A R24 10k
PG+14A R25 10k
PG3.3A R26 10k
PG3.3D R97 10k
Power Good
PowerGood1
GNDD4
U20B 7421
C32 0.1 uF

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
C24 0.1 uF
GNDD3
GNDA14 GNDA15
GNDA13
+3.3V +3.3VA −3V3 +15V −15V GNDA GNDD
PWR_FLAG

Buffer

S.WRITE_A S.LATCH_A
Pull up resistor To keep OE high (pins in high impedance state)
R22 10k
GNDD5
U6
SN74ALVCH16374DGGR
D−1…D−15 → 1Q1…1Q8 / 2Q1…2Q8 → A−1…A−15
A−[1..15]
TP_ADC_MSB1
+3.3V
C29 0.1 uF
JP6
+3V0 +3.3V
0.1 uF C31
GNDD1
S.CLK_A
TP_Clk_A1
+3V0
C44 0.1 uF
C45 0.1 uF
GNDA
C67 0.1 uF
GNDA6
U9 LTC2246HLX#PBF
CLK D0 LSB D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 MSB OF OE SHDIN MODE OGND OVDD VCM Ain+ Ain− REFH REFH REFL REFL SENSE
TP_Vcom1
C52 2.2 uF
R34 25
R35 25
TP_A+1
TP_A−1
C50 12 pF
C49 0.1 uF
C54 2.2 uF
C58 1 uF
C59 1 uF
+3V0
Vocm
AD8137YRZ U10
R43 1K
R44 1K
R49 1K
C30 2.2 uF
GNDA5
GNDA9
GNDA10
−14V0 +14V0 +3V0 GNDA
C53 C61 C65 C66
GNDA7
Clamping
D7
R58 500
R61 500
TP_Clamped1
TP_Range1
C74
R96
RV3
R95 2K
R94 1K
R64 500
Adjustable
5V
10V
20V
Gain Resistor/ Potentiometer
SW6
LT1363 U16
TP_Offset1
R66 4.99k
TP_Sum1
TP_In1
J8
GNDA11
GNDA12
Conn_01x04_Shielded J7
*note order of pins
Detector Video In
Offset voltage
−10V to 0V
0V to +10V
−10V to +10V
Presets
R27 402k
R90 402k
R28 143k
R89 143k
R91 221k
R92 221k
R93 8.87k
RV7
Adjustable
SW5

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